

Mohit Appari

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SUMMARY

Detail-oriented Data Scientist, Proficient Developer with 3+ years of experience applying statistical modeling, machine learning, and data engineering to solve complex problems in finance and analytics. With a strong foundation in mathematics and a passion for financial systems and Python, I've built scalable solutions for data-driven decision-making across business functions. I am eager to contribute to data science initiatives that bridge technical insights with strategic impact in high-stakes financial environments.

EDUCATION

- Florida State University, Masters in Data Science, Tallahassee, FL**

Aug 2023 - May 2025

 - Cumulative GPA: 4/4
 - Coursework: Statistics, Mathematics, Research, LLMs, Analytics, AI, Databases, Big Data, Business and Finance
- Christ University, Bachelors in Computer Science, Bangalore, IN**

Aug 2018 - May 2022

 - Cumulative GPA: 3.85/4
 - Coursework: Data Structures, Operating System, Software Architecture, Programming, Networking, Communication, SDLC

EXPERIENCE

- APD State of Florida, Data Scientist, Tallahassee, FL**

Jun 2025 - Present

 - Processed and **analysed 200K+** records across Medicaid, vendor, and client systems (serving **60K+ individuals**) by merging data from SQL Server, S3, and internal systems to generate unified datasets for reporting.
 - Led development** and maintenance of Power BI **dashboards** and automated **reports** using data pipelines built in **Python and SPSS**, for **executive leadership** and agency **staff**, supporting operational decisions and compliance tracking across Florida's disability services.
 - Optimized SQL queries** and data pipelines through indexing, partitioning, and caching strategies, reducing report execution time by **83%** (60 minutes to under 10 minutes).
 - Conducted **statistical analysis** using **SPSS, SAS, and R** to uncover billing anomalies, forecast financial trends, and deliver compliance insights, improving audit accuracy and budgeting processes. Led ad hoc and recurring data support for cross-functional teams, **reducing** manual workload by 40% through reusable code pipelines and enabling faster response to urgent data requests.
- Florida State University, AI/ ML Researcher, Tallahassee, Florida**

June 2024 - Sept 2024

 - Researched** open-source generative **AI** models, **algorithms (GANs, VAEs, RNNs)** to explore synthetic data generation for material property prediction and battery optimization applications.
 - Implemented and benchmarked NequIP (MIT Open Source) for molecular stability prediction using a dataset of **11,000+ materials**; improved prediction consistency by ~15% during initial validation.
 - Collaborated** with PhD researchers in Chemistry to align data science workflows with experimental research goals, enhancing interdisciplinary research on AI-driven material discovery and energy storage.
- S&P Global, Software Developer, Bangalore, India**

Jan 2022 - Aug 2023

 - Identified** the migration **opportunity** as an **intern**, auditing S&P's existing **.NET document-processing workflows** and evaluating **open-source alternatives** (regex, pdfplumber, Camelot, Tesseract, Textract) against licensed tooling.
 - Led a team of 5** through the migration, **owning the architecture** and **rollout** of a **Python-based extraction platform** that replaced the legacy system across multiple business domains, **cutting licensing costs** 80% and **improving extraction speed** 30%.
 - Built the pipeline** layer on **Cloud** (Docker, AWS) to process **10,000+** financial and automotive documents daily, with retry logic, structured logging, and fault-tolerant processing so the system ran unattended. **Reduced manual extraction time** by over 90%.
 - Utilized AI** in the workflow, combining **OCR with CNN-based classification** to turn unstructured PDFs and scanned documents into structured data, and worked with the team to **improve entity recognition accuracy** on financial documents
 - Engineered the platform for long-term maintainability with modular reusable components, CDC-based incremental updates, and 80% test coverage, backed by GitHub Actions CI/CD for reliable releases.
- LG Electronics, Software Developer Intern, Bangalore, India**

May 2021 - Aug 2021

 - Built a **real-time drowsiness detection system** using **Python, OpenCV, YOLO**, and Raspberry Pi, achieving 90% facial landmark accuracy to improve driver safety in logistics operations.
 - Designed a CNN-powered logistics tracking system** integrated with REST APIs and PostgreSQL, reducing package sorting time by 25% and earning **3rd place** in LG's internal **Ideathon** competition.
- ITC Limited, Software Developer Intern, Bangalore, India**

Jan 2021 - Mar 2021

 - Developed** a master data management system using **Python** (Flask, REST API) and **MySQL** with multi-role access control, implementing role-based authentication and approval workflows to improve data security and user efficiency.
 - Automated** product tracking and data update **processes, reducing manual workload** by **30%** and enhancing data accuracy across multiple business units.

PROJECTS

- BayWheels Lyft Trip Analytics** | Python, Prefect, Apache (Spark, Kafka), Docker, Terraform, GCP (BigQuery, GCS), dbt
 - Built** a complete **data pipeline** to process over **400K+ monthly bike ride records** using Python, Prefect, Docker, and GCP. Used Terraform to set up cloud infrastructure, dbt for data modelling, and BigQuery for analysis and storage. Used Kafka and Spark to handle both batch and real-time data, making it easy to track ridership trends, station usage, and trip patterns. **Cut down manual work by 90%** and made reports 60% faster.
- STA5635 RAG App** | Python, LangChain, OpenAI, Cohere, Pinecone, Flask, FastAPI, Streamlit, Docker, GitHub, Vercel
 - Built a **high-performance RAG** application using LangChain, OpenAI, Cohere, Pinecone, Flask, and FastAPI, enabling contextual Q&A over academic PDFs. **Reduced research lookup time by 70%** through real-time document retrieval and LLM-powered summarisation, improving user efficiency and insight accessibility

SKILLS

Programming: Python, R, SQL, Rust, C, C++, Java, JavaScript, TypeScript, HTML, CSS, Bash, Linux, PySpark, Go, Scala
Databases: PostgreSQL, MySQL, Oracle, MongoDB, Neo4j, Firebase, Redis, Cassandra, DuckDB, Snowflake, BigQuery, Delta Lake, Pinecone,
Cloud: AWS (EC2, S3, Lambda, RDS, SageMaker), GCP (BigQuery, Vertex AI), Azure Databricks, Kubernetes, Docker
Frameworks: PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, Prophet, Flask, FastAPI, Django, Streamlit, Dash, React, Node.js
Other Tools: Github, MLflow, Airflow, dbt, Kafka, Spark, Hadoop, Terraform, n8n, OpenAPI, Cohere, Ollama, Cursor, Langchain, Langgraph, Codex